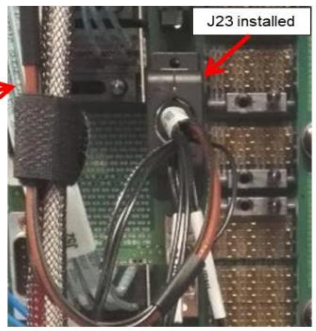
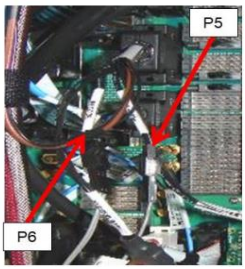
Build Overview:

1 **Ethernet Switch Install**

2 **Install the following cables:**

- **a.SB Sync**
- **b.CLK Fanout**
- **c.New cable**
- **d.TDR Coax**
- **e.CLK100 Loopback**
- **f.CD Cal Bus**
- **g.HV Interlock**
- **h.DRG Suppor**





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED TECHNICIANS.

MATERIAL:

NO.	PART NO.	DESCRIPTION	QTY.
1	655-378-00	ELECTRO MECHANICAL, UF-II TESTHEAD ETHERNET SWITCH SYSRS, FOR SEMI TEST	1
2	633-782-02	ALUMINUM RAIL ETHERNET SWITCH BRACKET OR MISC ARTICLE	2
3	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE WASHER. O.D. 11MM	5
4	627-718-02	CABLE DRG SBDB2 SYNCX TELECOM WITH CONN	4
5	813-929-01	HDW VELCRO STRP 5/8	1
6	626-253-00	CABLE UF_CLKFANOUT_24 LCXCEDE TELECOM WITH CONN	1
7	627-083-00	CABLE UF2 TDR COAX TELECOM WITH CONN	4
8	813-929-01		1
9	650-679-00	CABLE UF2 12SLOT TH_INT_SMCLON_PRESS TELECOM WITH CONN	1
10	659-442-02	CABLE UF2 CLK100 TH BULKHEAD W SOLDER TELECOM WITH CONN	1
11	633-807-00	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
12	633-807-01	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
13	633-807-02	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
14	633-807-03	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
15	627-191-04	CABLE DRGSPT4 TO DBMISC 26 TELECOM WITH CONN	1
16	627-191-03	CABLE DRGSPT3 TO DBMISC 26 TELECOM WITH CONN	1
17	627-191-02	CABLE DRGSPT2 TO DBMISC 26 TELECOM WITH CONN	1
18	627-191-01	CABLE DRGSPT1 TO DBMISC 26 TELECOM WITH CONN	1
19	655-265-01	CABLE BACLE_HV_INTLK_INT TALL CONN TELECOM WITH CONN	1

METHOD:

Flow chart of operation steps

BPM preparation
↓
Preparation Tools
↓
Prepare Materials
↓
Complete Operation
↓
Inventory Materials
↓
Operation Check
↓
Inventory tool

MACHINE:

TOOL NAME	TORQUE VALUE
PH2 PHILLIPS TORQUE SCREWDRIVER	0.7 N.m
PH2 PHILLIPS TORQUE SCREWDRIVER	3.3 N.m
FLAT-BLADE SCREWDRIVER	0.19 N.m
TORQUE SCREWDRIVER	0.7 N.m
14MM SOCKET	0.7 N.m
3/16 SOCKET WRENCH	0.19 N.m
OPEN-END WRENCH	NA

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
1	655-378-30	ELEC-MECH, UF2 PREASSEMB/PREPRGRM 10G TESTHEAD ENET SWITCH	1
2	633-782-02	ALUMINUM RAIL ETHERNET SWITCH BRACKET OR MISC ARTICLE	2

MACHINE: N/A

METHOD:

Operation Steps

1.1 Install the network switch and take the following materials:

a. Take 1 Ethernet switch (655-378-00).

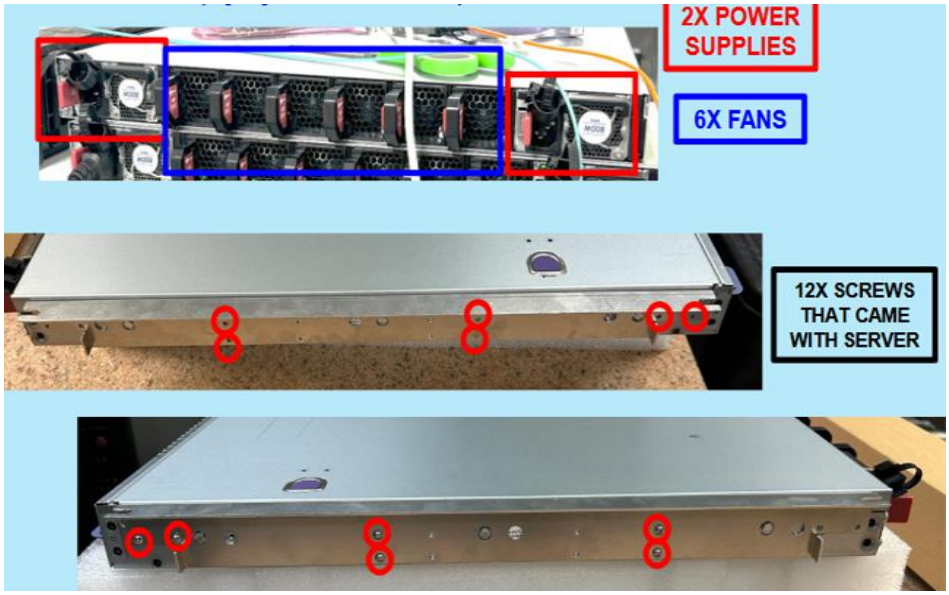
b. Take 2 brackets (633-782-02).

c. Take (x12) M4 screws supplied with the switch. (From Supplier)

d. Orient the switch as shown below. Note the location of the power supplies, rear left.

e. Attach Mounting Brackets 633-782-06 (x2) to the switch using 6 M4 screws per side that came with the switch. Torque screws to .7 N•m.

Note: Make sure the flange of the bracket is on the table surface.



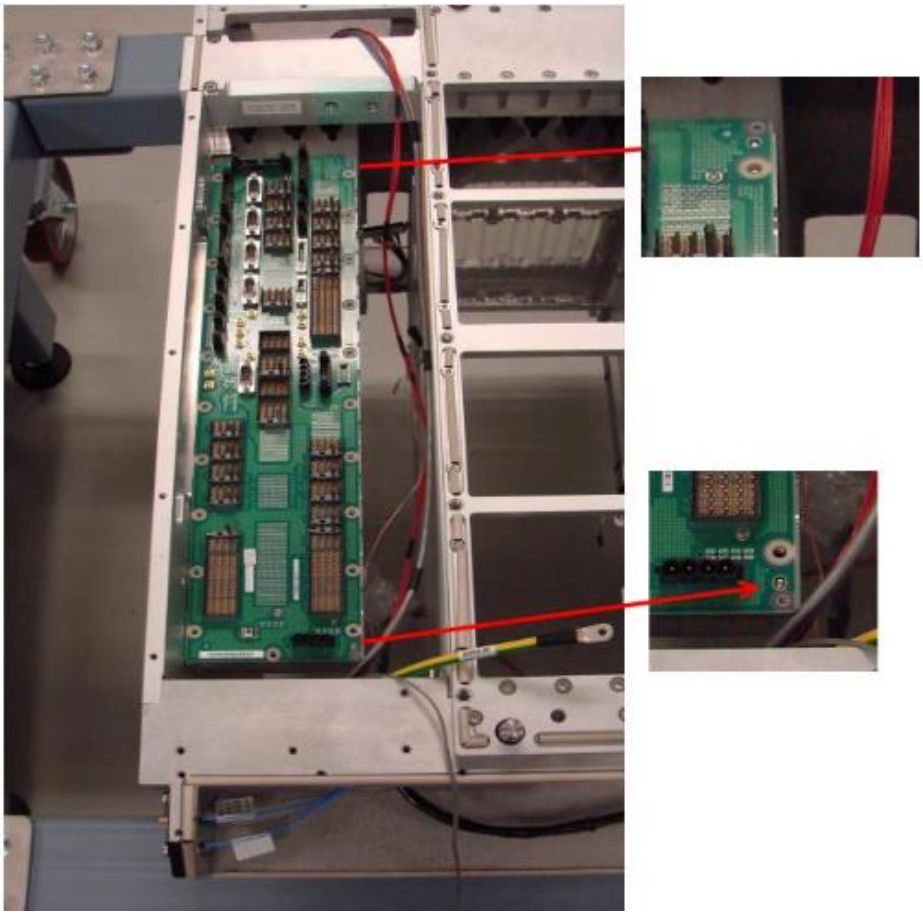
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

6.3 Place the backplane into the T/H frame. Note that the two positioning pins in the frame must be inserted into the two positioning holes of the backplane.



MAN: CERTIFIED OPERATOR, TECHNICIAN

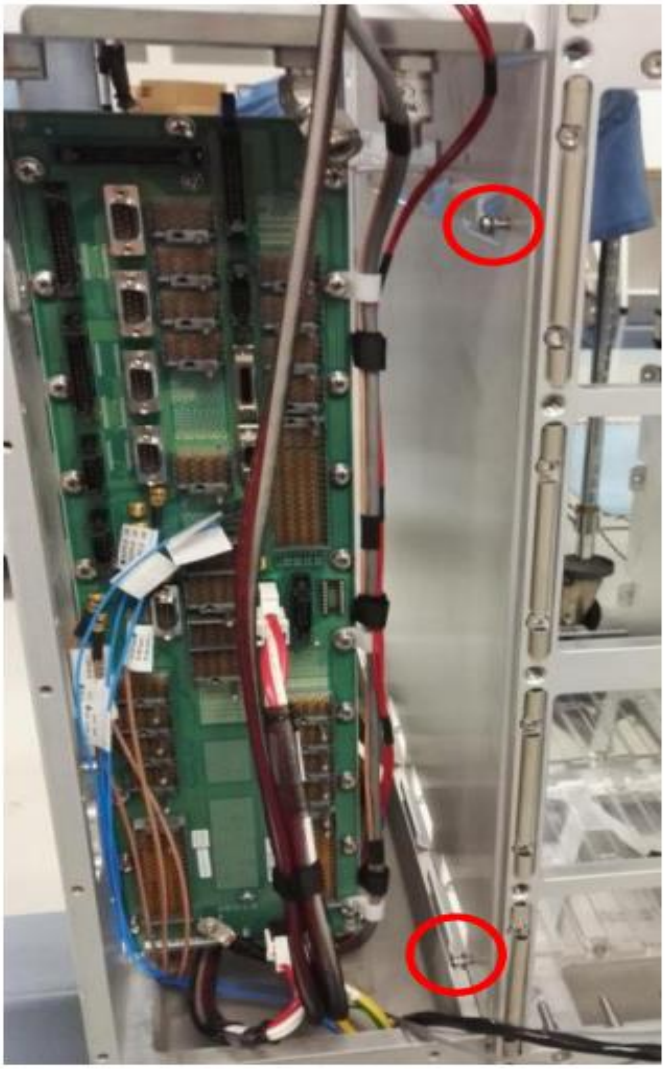
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
3	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X.0x12MM L W/CONE W	1

MACHINE: N/A

METHOD:

1.4 Install 2 screws (472-426-12) on the back panel as shown in the figure above and below the T/H frame.
Install the screws but do not tighten them.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
4	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8x12MM L W/CONE	2

MACHINE: PH2 Phillips torque screwdriver 3.3 N·m.

METHOD:

1.5 Install the network switch into the T/H frame as shown in the figure, and ensure that the bracket on the switch is inserted into the previously assembled screws.

1.6 Install 2 screws (472-426-12) on the front of the switch as shown in the figure.
The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.

1.7 Use a tool to fix the two screws on the back with a PH2 Phillips torque screwdriver. The torque value is 3.3 N·m.

1.8 When the switch is installed in the T/H frame as shown in the figure, you must ensure that the two sockets on the switch are at the bottom.

Use Prolong PH2 with Milwaukee 90°

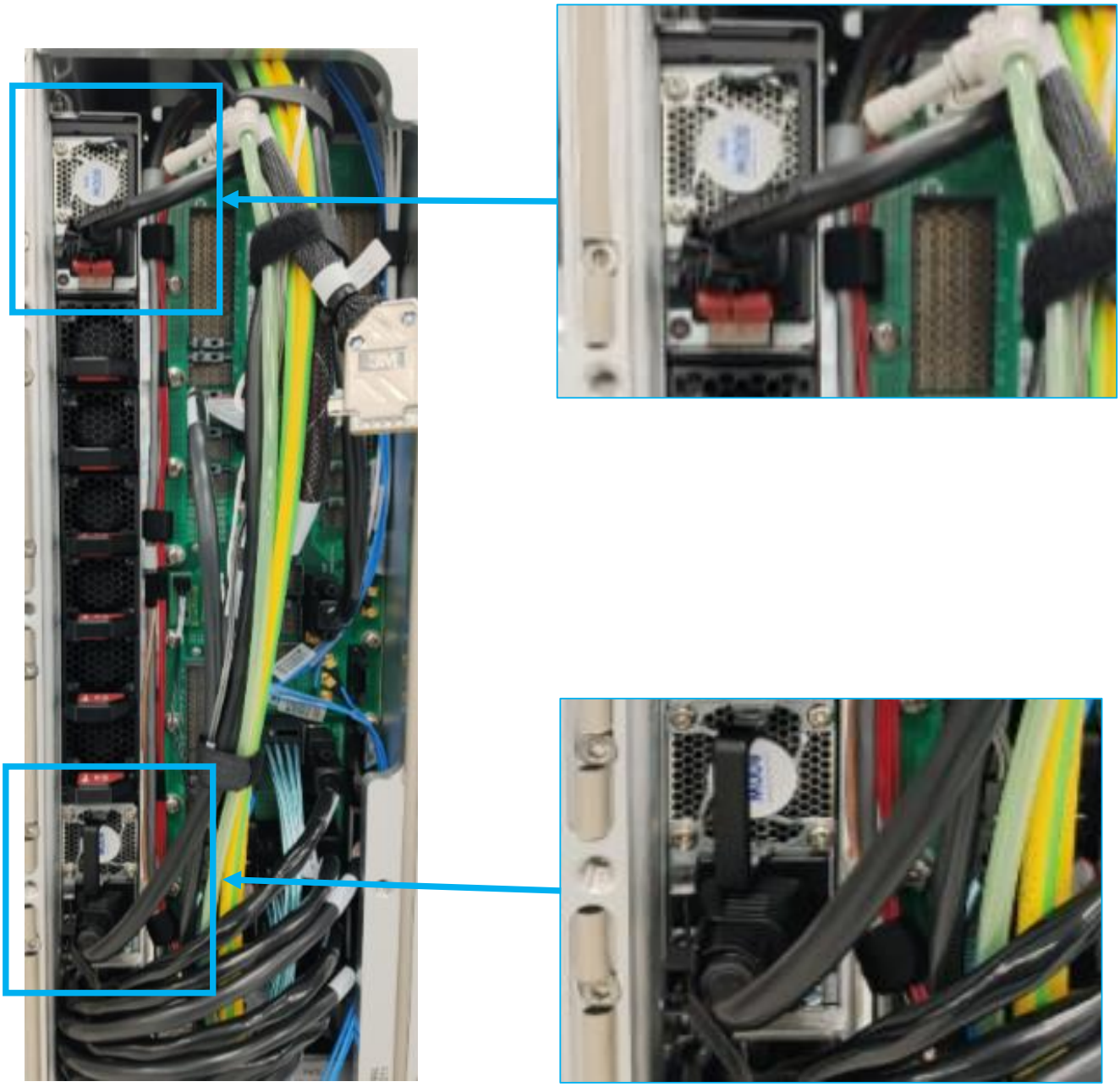
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

1.9 Insert the two power cord plugs (710-502-55) into the back of the switch as shown in the figure.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
5	627-718-02	CABLE DRG SBDB2 SYNCX TELECOM WITH CONN	4

MACHINE: N/A

METHOD:

2.1 Install 4 DRG SBDB2 syncx line(627-718-02).

2.2 Label the lines in the following order:

a. Mark "J58" on the backplane side and "DB3_J4" on the connector side.

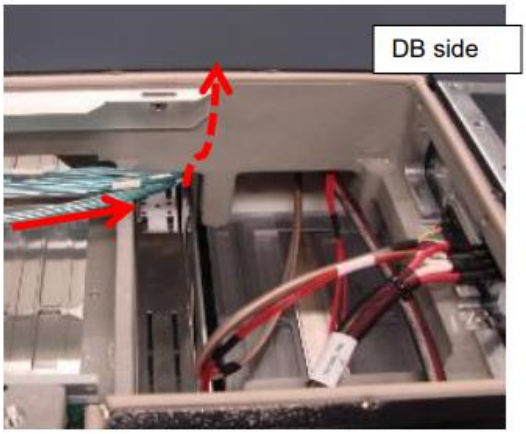
b. Mark "J57" on the back panel end and "DB1_2J14" on the connector end.

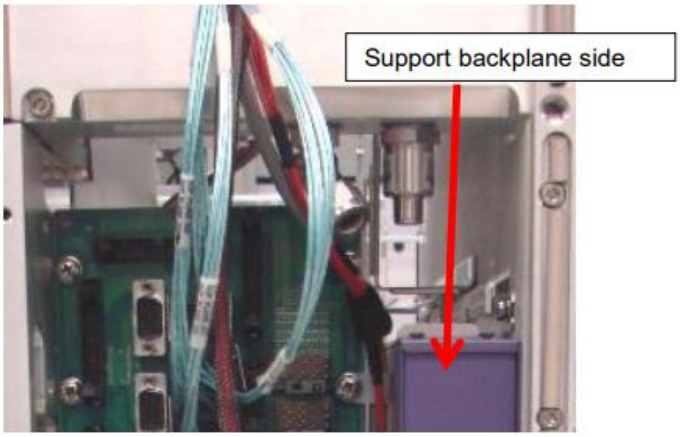
c. Mark "J56" on the backplane side and "DB1_2J4" on the connector side.

d. Mark "J55" on the backplane end and "DB_0J4" on the connection head end.

Label markings " SUP PLANE J58" and DB3_J4".
Repeat for the other three cables as noted.

2.3 As shown in the T/H direction, pass the 4 wires at one end of J55-J58 through the T/H opening, and then place the wires on the back plate end.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

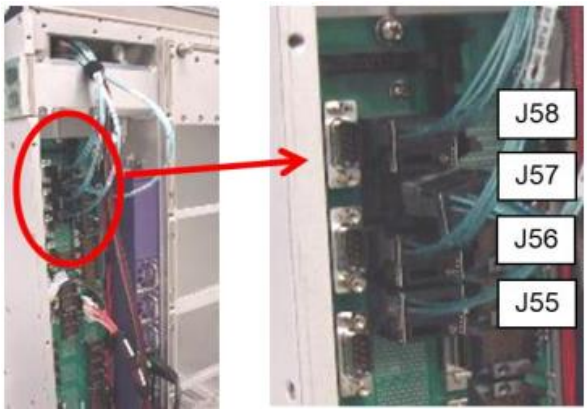
ITEM	PART NUMBER	DESCRIPTION	QTY
6	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE: N/A

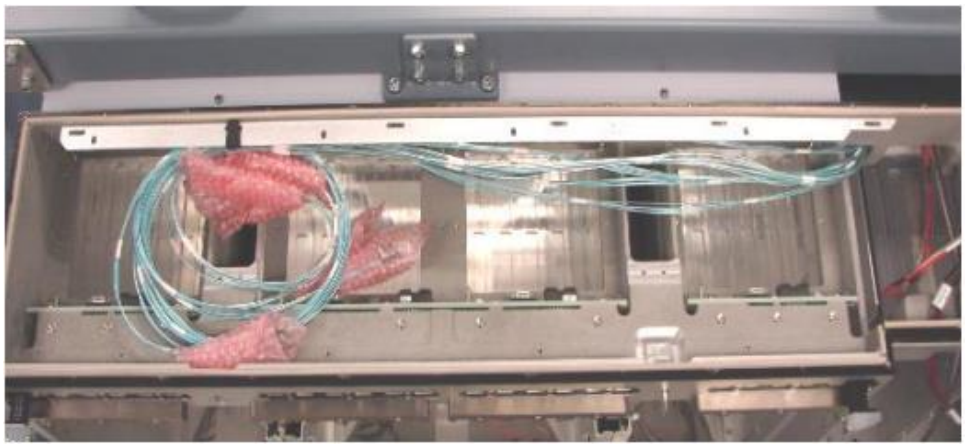
METHOD:

2.4 Insert the labels J58, J57, J56 and J55 on the back panel.

2.5 Tighten the wire with the existing black cable tie.



2.6 Place the 4 wires as shown in the figure, encircle the wires and use a tie (813-929-01) to fix the wires on the T/H frame.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
7	629-253-00	CABLE UF_CLKFANOUT_24 LCXCEDE TELECOM WITH CONN	1

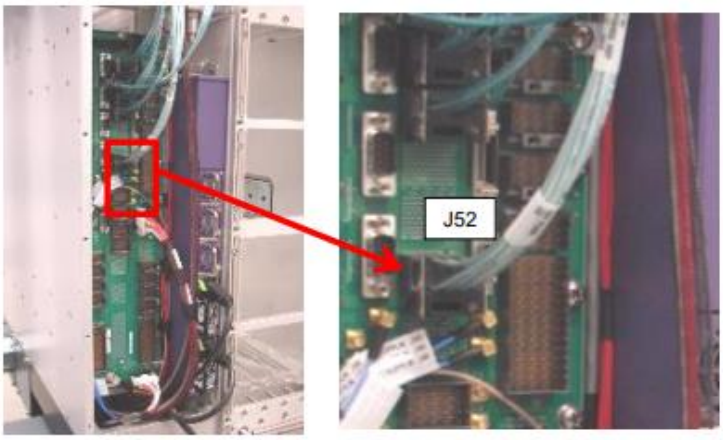
MACHINE: N/A

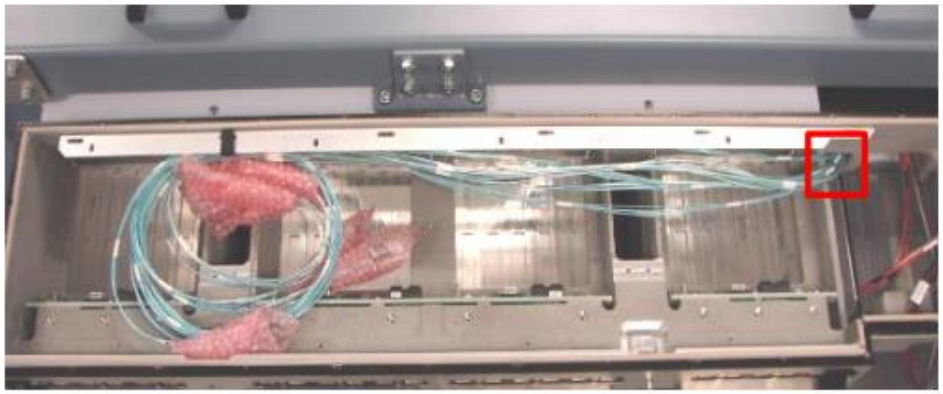
METHOD:

3.1 Install the CLK FANOUT line (629-253-00).

3.2 Insert the connector labeled "J52" on the line into the backplane end.

3.3 Thread the wire into the T/H frame slot as shown in the figure, place the wire in a circle like the previous 4 wires, and tie the wire tightly with the same cable tie.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
8	627-083-00	CABLE UF2 TDR COAX TELECOM WITH CONN	4

MACHINE: N/A

METHOD:

4.1 Install 4 TDR COAX lines (627-083-00).

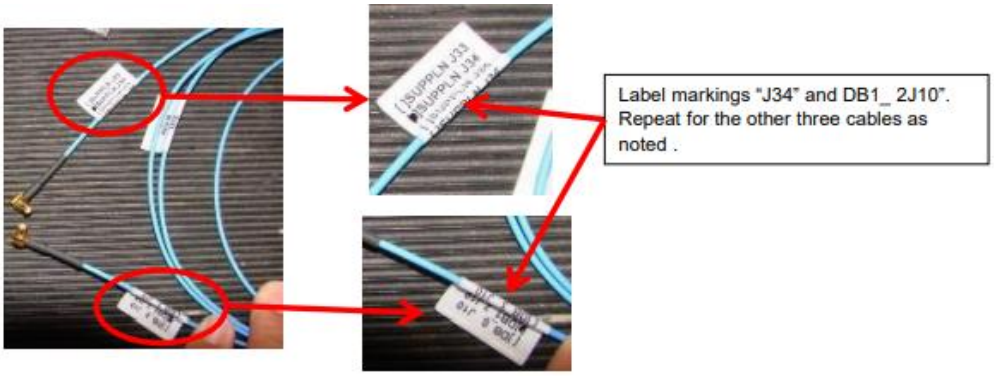
4.2 Label the lines in the following order:

a. Mark "J33" on the back panel end and "DB_0 J10" on the other end.

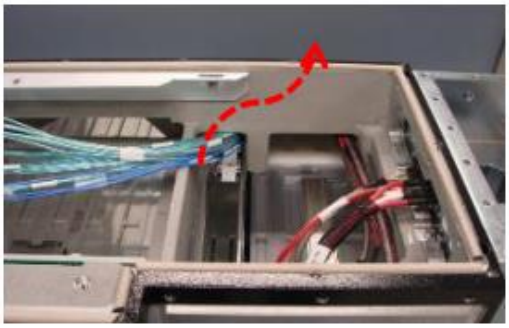
b. Mark "J34" on the back panel end and "DB1_2 J10" on the other end.

c. Mark "J35" on the back panel end and "DB1_2 J11" on the other end.

d. Mark "J36" on the back panel end and "DB_3 J10" on the other end.



4.3 As shown in the figure, insert the wire label "J33-J36" connector into the opening slot, and put it on the back plate end through the T/H frame channel.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

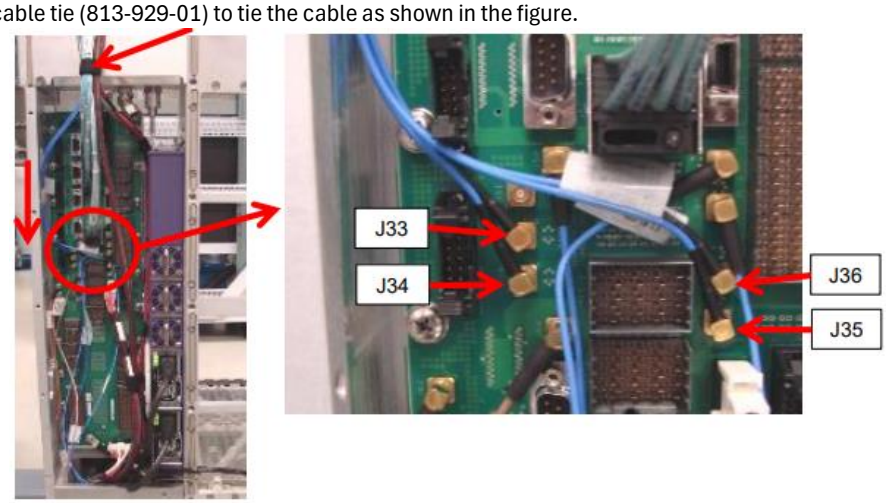
ITEM	PART NUMBER	DESCRIPTION	QTY
9	813-929-01	HDW VELCRO STRP 5/8	1

MACHINE: N/A

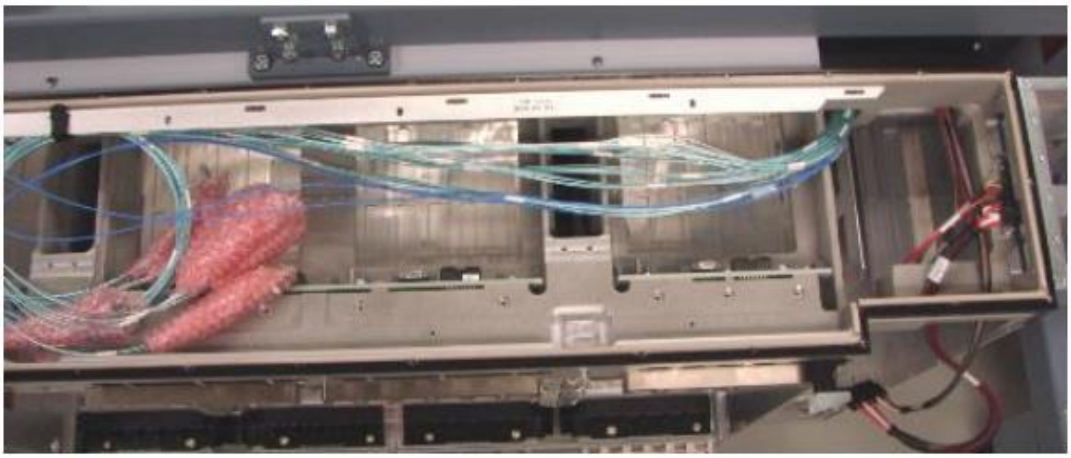
METHOD:

4.4 Route the wires along the edge of the T/H frame and insert them into the back panel ends J36, J35, J34 and J33 according to the labels.

4.5 Use the black cable tie (813-929-01) to tie the cable as shown in the figure.



4.6 Place the wire together with the existing harness as shown.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
10	650-679-00	CABLE UF2 12SLOT TH_INT_SMCLON_PRESS TELECOM WITH CONN	1

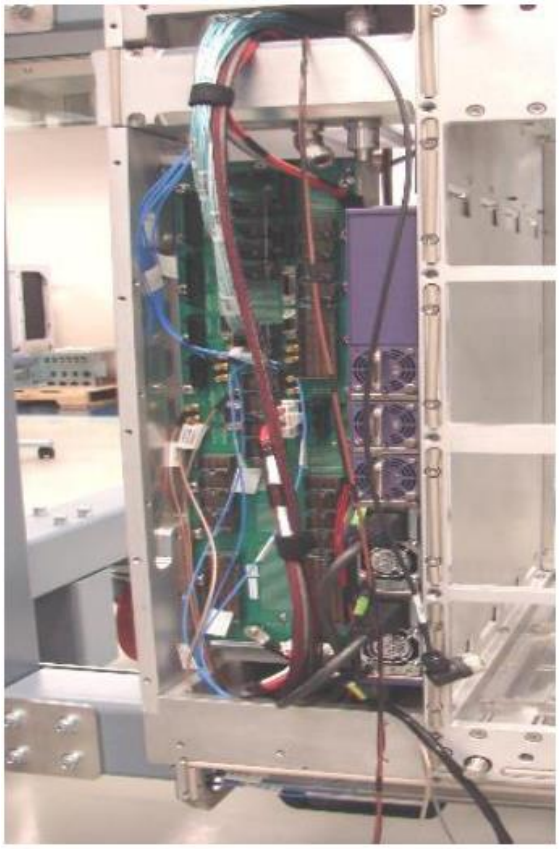
MACHINE: N/A

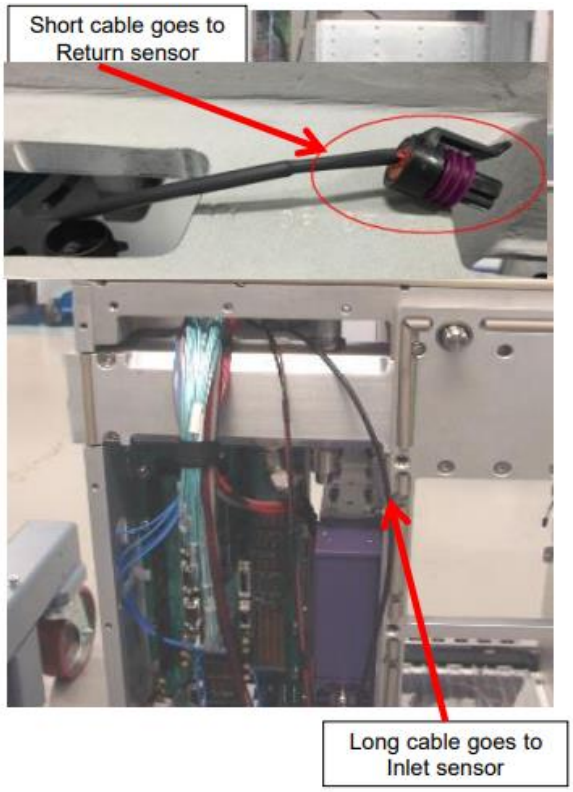
METHOD:

5.1 Install one CLK100 LOOPBACK line (650-679-00).

5.2 As shown in the figure, From the Support backplane side, route J1 connector across the cable channel to the DB side. Place J1 connector near the disconnect panel area. It will get installed to the disconnect panel in a later step.

5.3 Split the two pressure sensor wires. The long wire stays on the support backplane side near the return pressure sensor and the short one routes inside.





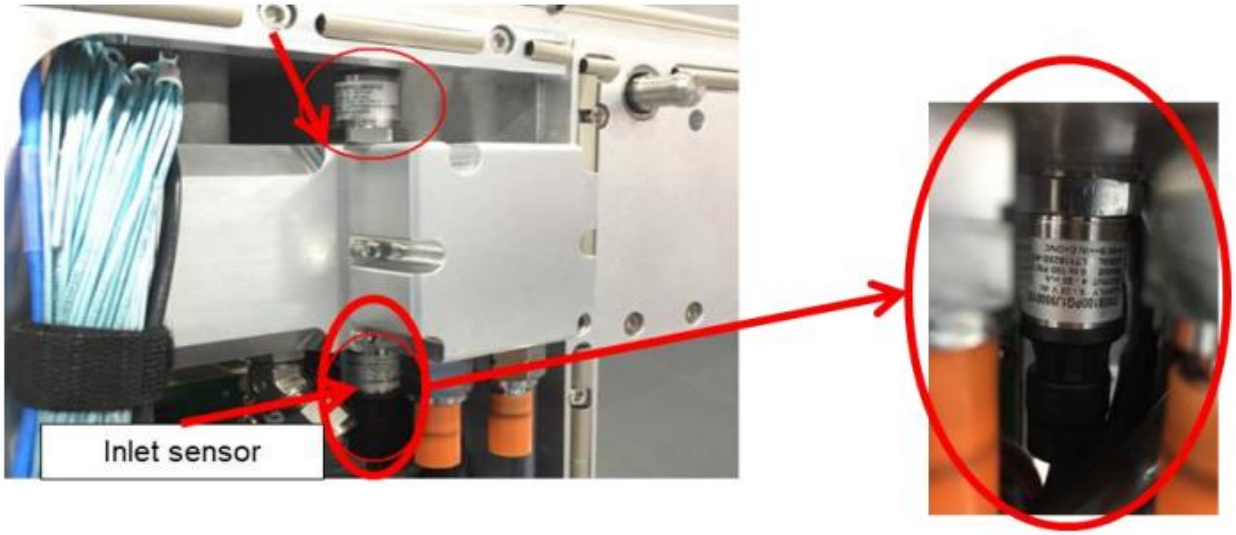
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

5.5 Place the long pressure sensor wire under the Manifold as shown, facing the back of the pressure sensor.
Label the line "Press_1 Inlet". Route the long pressure sensor cable down over manifold block and towards the back of the pressure sensor.
The connector is inserted into the inlet pressure sensor according to the position of the buckle. After the card is locked,
confirm whether it is in place.



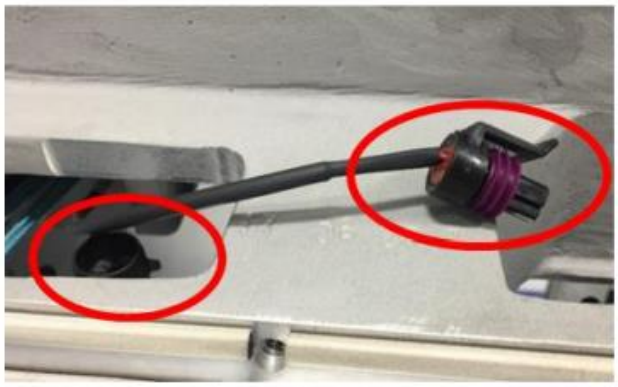
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

5.6 Place the short pressure sensor wire above the Manifold as shown in the figure, towards the back of the pressure sensor.
Add the line label "Press_2 Return"
The connector is inserted into the inlet pressure sensor according to the position of the buckle.
After the card is locked, confirm whether it is in place.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

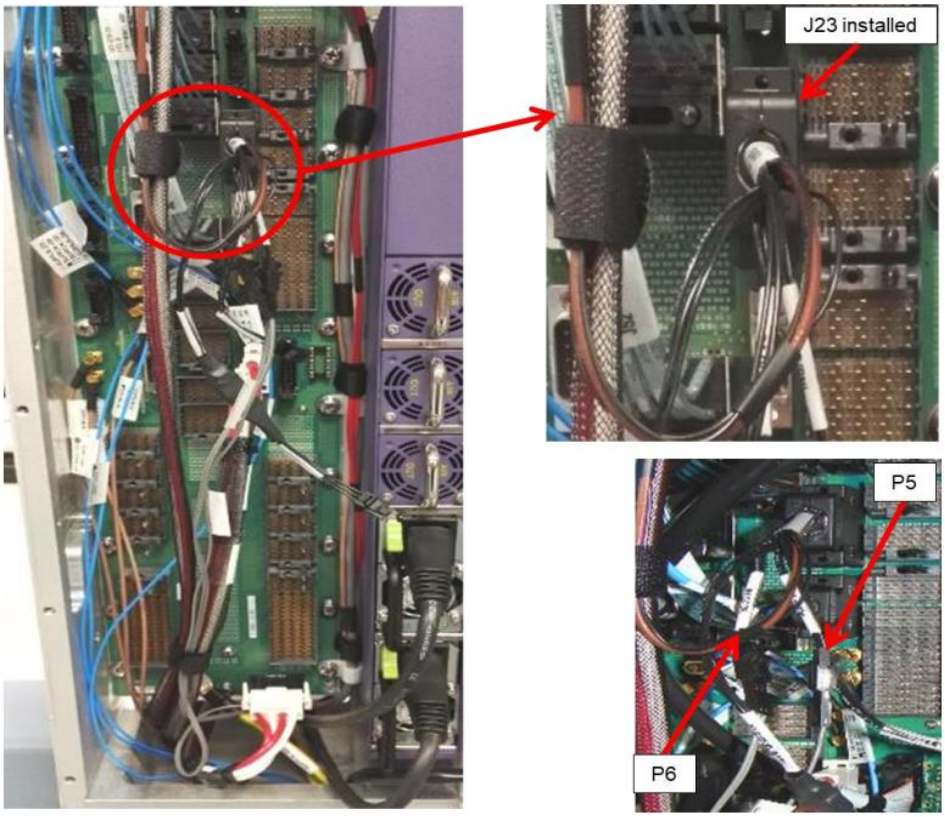
METHOD:

5.7 Insert the big black wire connector into the "J23" interface on the back panel according to the label as shown in the figure.

5.8 Route the wires according to the existing harness as shown in the figure, and use the existing cable ties to tighten the wires.

5.9 After installing the J23 cable, insert the P5 and P6 connectors into the key switch and the door lock switch respectively.

i.e. SUPPLN J23-P5, SUPPLN J23-P6



Safety Tips: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
11	659-442-02	CABLE UF2 CLK100 TH BULKHEAD W SOLDER TELECOM WITH CONN	1

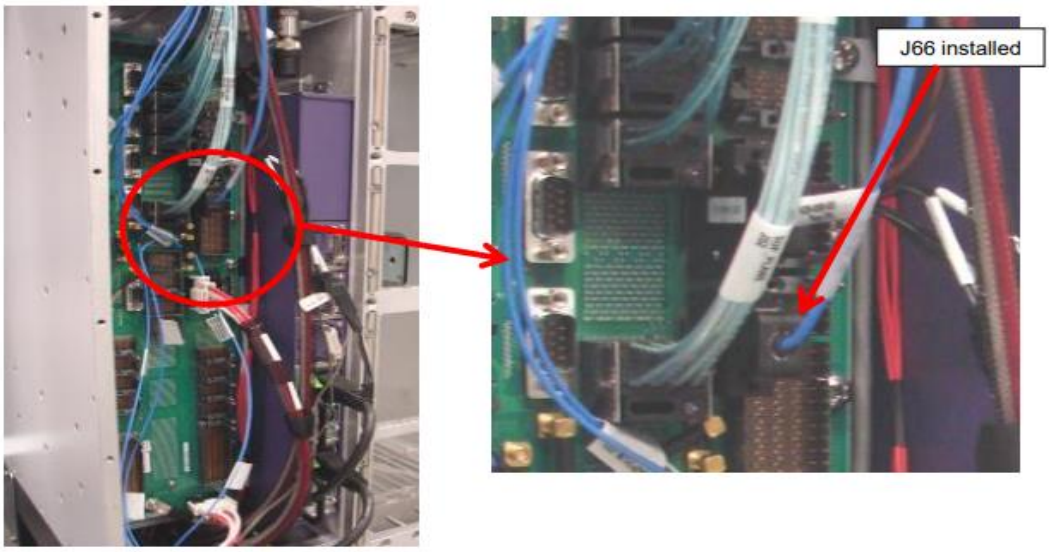
MACHINE: N/A

METHOD:

6.1 Install the CLK100 TH Bulkhead cable (659-442-02).

6.2 Install the black plastic connector to the "J66" end of the backplane according to the cable label.

6.3 Place the J4 connector of the cable through the T/H frame channel on the opposite panel as shown in the figure. Install the J4 interface later.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

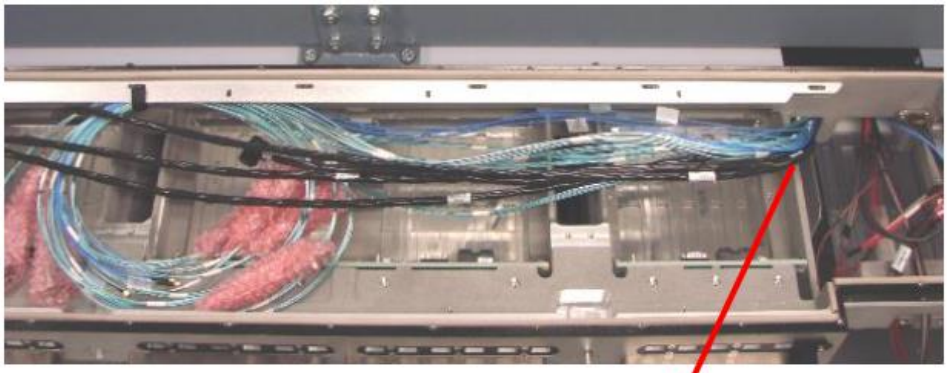
ITEM	PART NUMBER	DESCRIPTION	QTY
12	633-807-00	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
13	633-807-01	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
14	633-807-02	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1
15	633-807-03	CABLE UF2 DC_CAL_BUS 75 V TELECOM WITH CONN	1

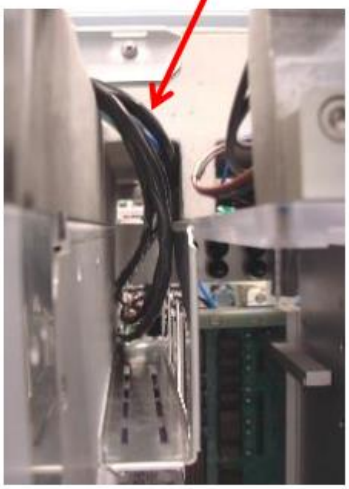
MACHINE: N/A

METHOD:

6.1 Install the DC_CAL_BUS, 75V line (633-807-00, 633-807-01, 633-807-02, 633-807-03).

6.2 Pass the thread into the T/H opening slot as shown in the figure, and place it on the backplate end through the shelf inside the T/H.





MAN: CERTIFIED OPERATOR, TECHNICIAN

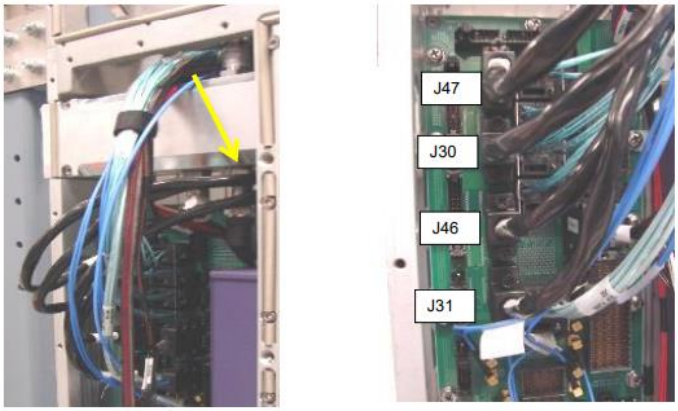
MATERIAL: N/A

MACHINE: flat-blade screwdriver 0.19 N·m

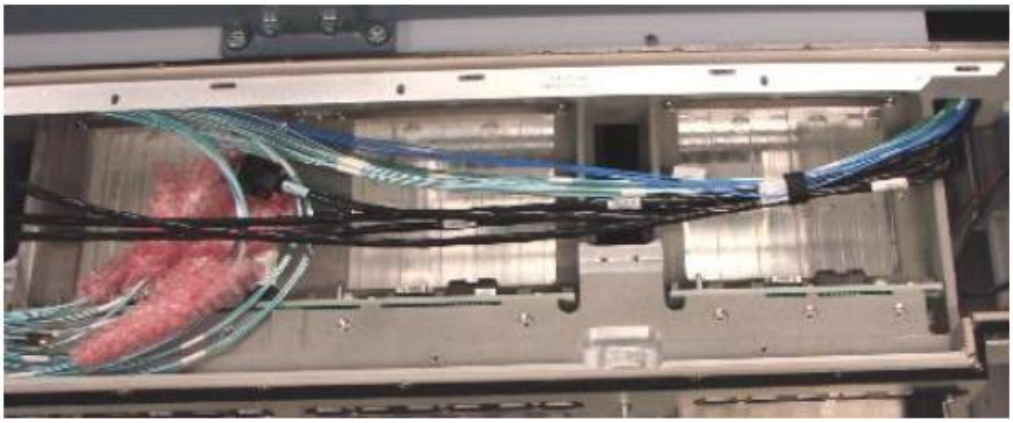
METHOD:

6.3 Route these 4 wires as shown in the figure.

6.4 As shown in the figure, insert these 4 wires into the back plate end according to the labels J47, J30, J46, and J31, and fix them with a flatblade screwdriver. The torque value is 0.19 N·m.



6.5 Place the 4 black wires as shown in the figure and tie the wire harness with the existing cable tie.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
16	627-191-04	CABLE DRGSPT4 TO DBMISC 26 TELECOM WITH CONN	1
17	627-191-03	CABLE DRGSPT3 TO DBMISC 26 TELECOM WITH CONN	1
18	627-191-02	CABLE DRGSPT2 TO DBMISC 26 TELECOM WITH CONN	1
19	627-191-01	CABLE DRGSPT1 TO DBMISC 26 TELECOM WITH CONN	1

MACHINE: N/A

METHOD:

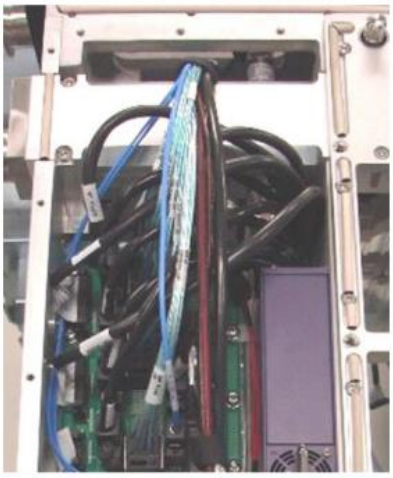
7.1 Install DRGSPTX TO DBMISC 26 cables (627-191-04, 627-191-03, 627-191-02, 627-191-01).

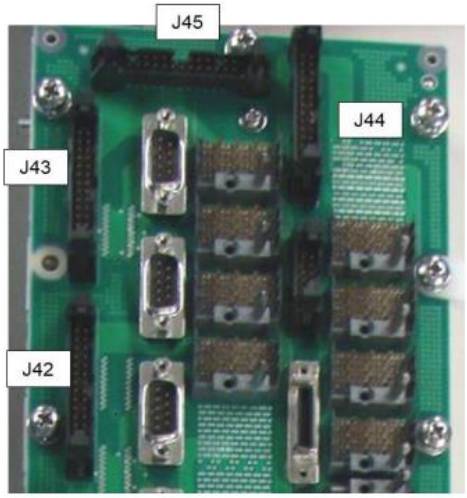
7.2 Insert the thread into the T/H open slot as shown in the figure, and place it on the back plate end through the rack inside the T/H.

7.3 Route and cross these 4 lines between the previous black lines as shown in the figure.

7.4 Insert these 4 wires into the backplane end according to the labels J45, J44, J43 and J42 as shown in the figure.

7.5 Place these 4 black wires and tie the wire harness with the existing cable tie.





MAN: CERTIFIED OPERATOR, TECHNICIAN

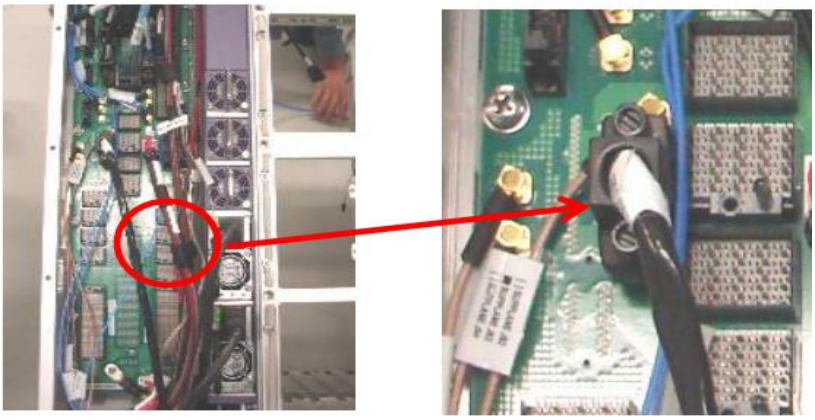
MATERIAL: N/A

MACHINE: flat-blade screwdriver 0.19 N·m

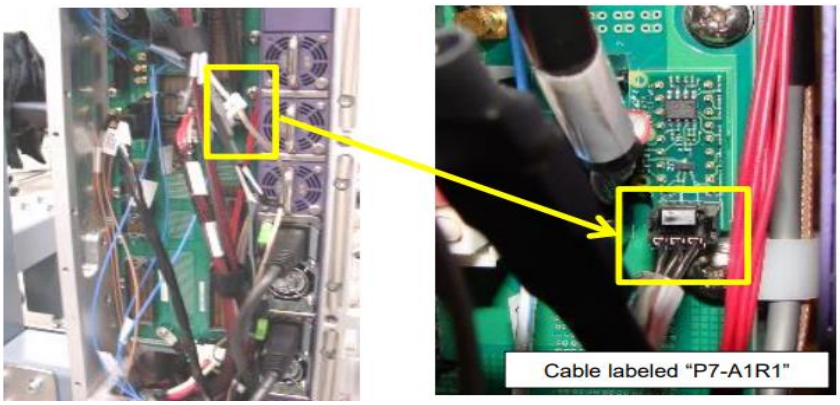
METHOD:

8.1 Install the Interlock cable and NVRAM PCBA.

8.2 Install the interface of the previously assembled Interlock cable 626-232-00 to the J32 interface on the backplane.
Use a flat-blade screwdriver to fix the torque value is 0.19 N·m.



8.3 Install the cable connector labeled "P7 SYSNVRAM".



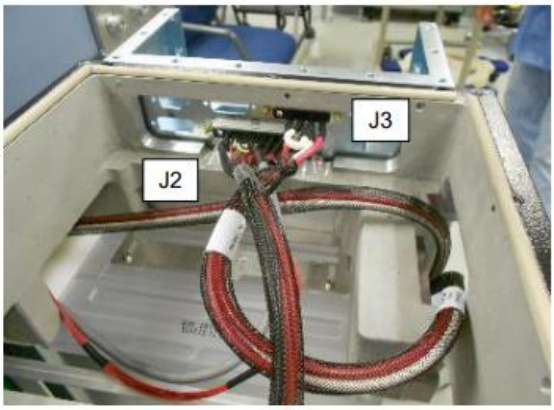
MAN: CERTIFIED OPERATOR, TECHNICIAN

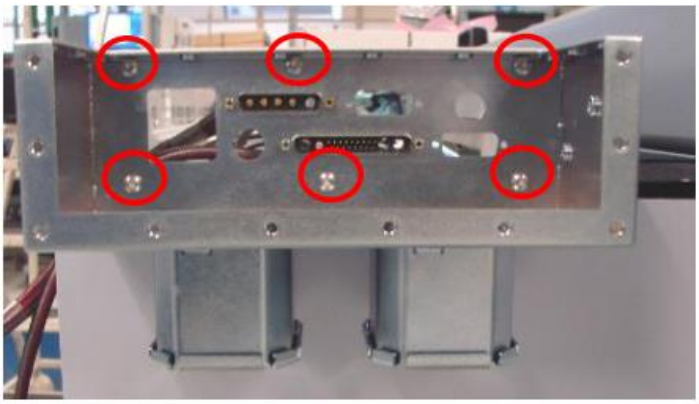
MATERIAL: N/A

MACHINE: N/A

METHOD:

9.1 Loosen the six M5 screws on the panel as shown, and remove the cover. Note that there are J2 and J3 connectors on the panel. Save these 6 M5 screws to be used in subsequent assembly.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
20	655-265-01	CABLE CABLE_HV_INTLK_INT TALL CONN TELECOM WITH CONN	1

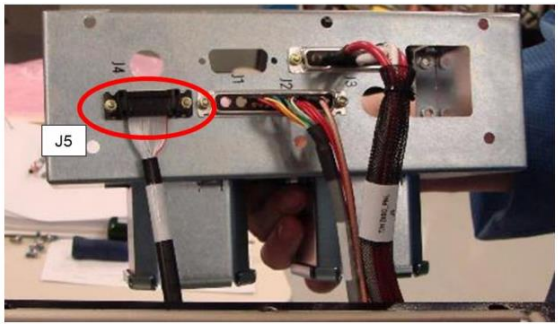
MACHINE: 3/16 socket wrench 0.19 N.m

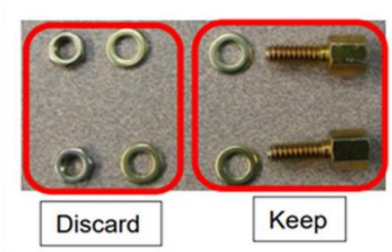
METHOD:

10.1 Install 1pcs HV_INTLK_INTERNAL cable (655-265-01).

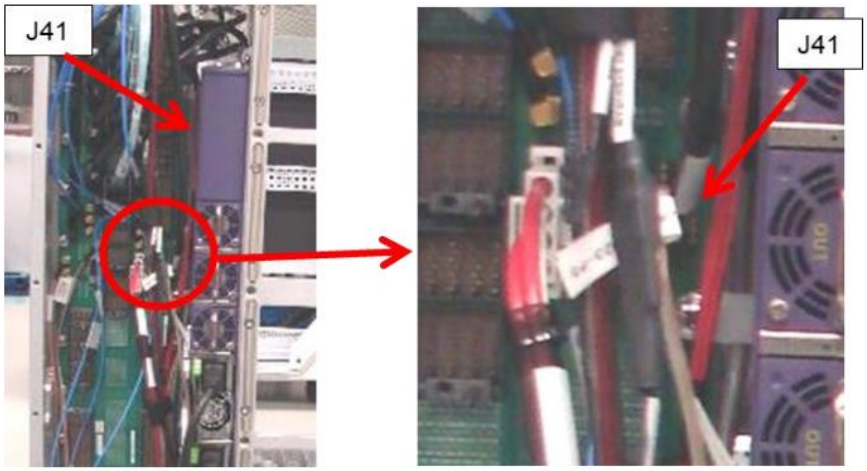
10.2 Remove hardware from connector J5. Save 2 sets of hex nuts and nut posts. Discard extra split-lock washers.

10.3 Use two sets of hexagon nuts and studs to fix the J5 connector on the panel, the tool is a 3/16 socket wrench, and the torque value is 0.19 N·m.





10.4 Place the other end of the cable on the backplane end, insert the EMO cable according to the connector label "J41"



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: 3/16 socket wrench 0.19 N·m

METHOD:

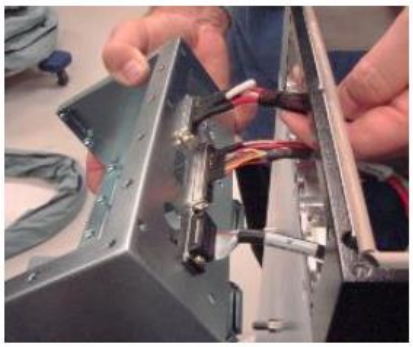
11.1 Install the CLK100 LOOPBACK cable (650-679-00) and CLK100 TH cable (659-442-02) on the panel.

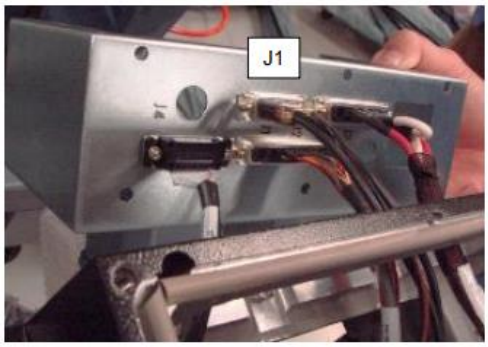
11.2 For cable 650-679-00:

a. Remove and retain hardware from cable connector J1. Save it and use it for subsequent assembly.

b. Use 2 sets of nut posts, spring washers and hex nuts to fix the J1 interface on the panel.

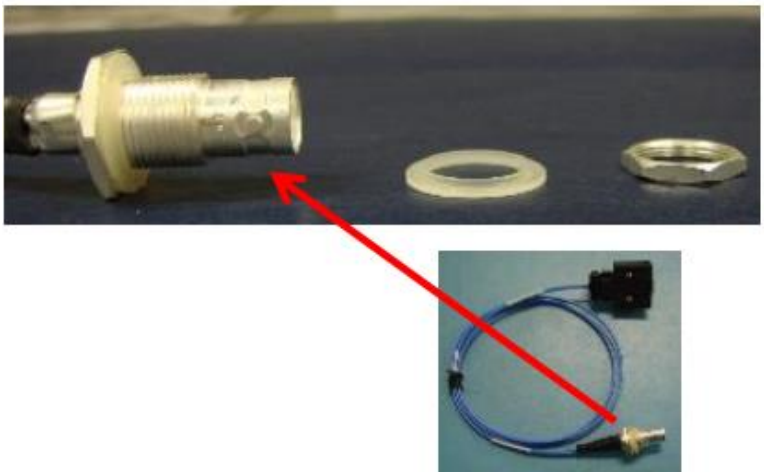
c. Use a tool to fix the 3/16 socket wrench with a torque value of 0.19 N·m.





11.3 For cable 659-442-02:

a. Remove the nut on the wire connector and an adjacent plastic washer, save it and use it for subsequent assembly.



Safety Tips: N/A

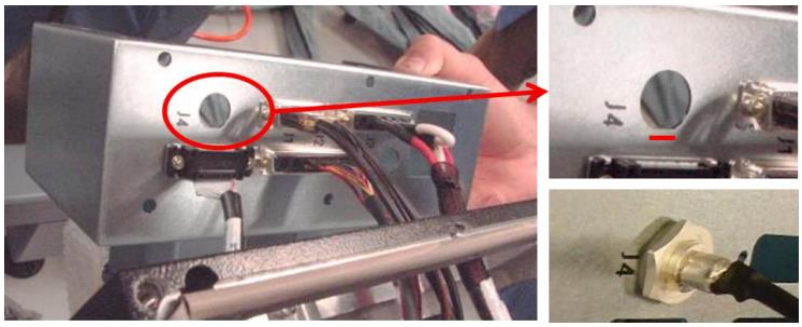
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

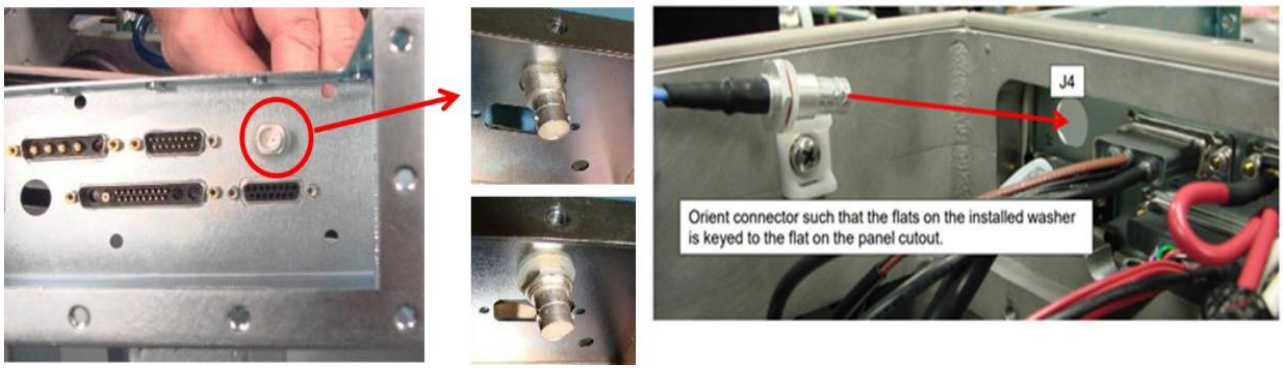
MACHINE: N/A

METHOD:

b. Pass the wire connector through the T/H opening slot, and install connector J4 on the panel. When installing, ensure that the plane on the connector head is placed on the plane on the panel.
When rotating the wire, make sure not to drop the gasket on the connector.



c. From the inside of the disconnect panel attach connector J4 by installing the shoulder washer and hex nut saved from before.
Orient the shoulder washer with the flat matching the flat on the panel cutout as before.



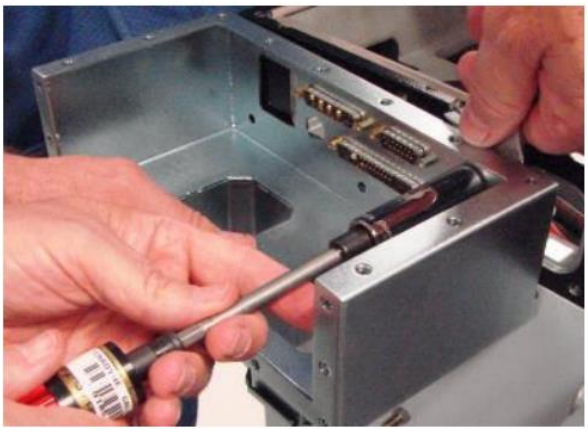
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

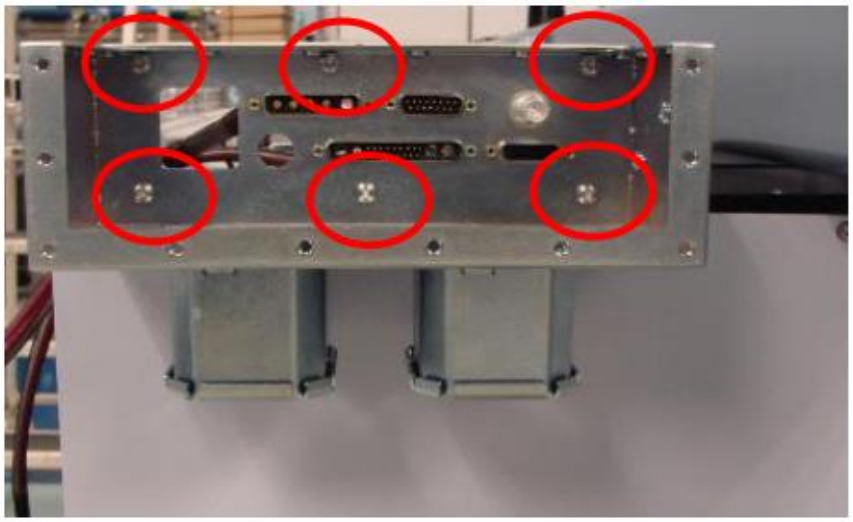
MACHINE: 14mm socket 0.7 N·m, open-end wrench, PH2 Phillips torque screwdriver 3.3 N·m.

METHOD:

d. Use a tool 14mm socket and open-end wrench to fix, and the torque value is 0.7 N·m.



11.4 Re-fix the panel on the T/H frame with 6 M5 screws that were removed before.
The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.



Safety Tips: N/A

TQC: N/A

MAN: CERTIFIED OPERATOR, TECHNICIAN

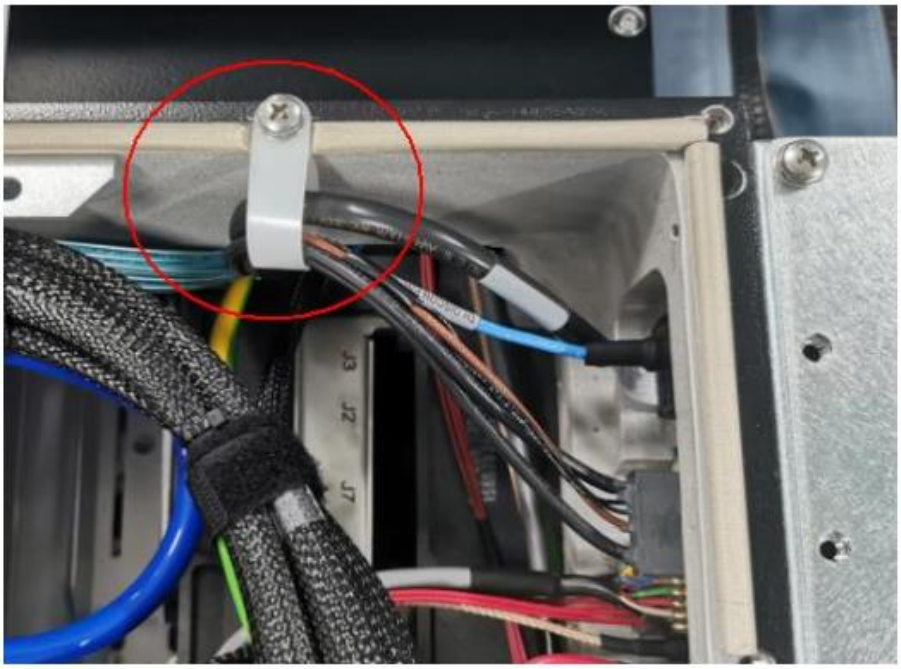
MATERIAL:

ITEM	PART NUMBER	DESCRIPTION	QTY
21	835-610-42	BUCKLE CLAMP	1
22	472-426-12	PAN HEAD CROSS RECESSED SCREW M5X0.8 x 12MM L W/CONE	1

MACHINE: PH2 Phillips torque screwdriver, 1.6 N·m

METHOD:

11.5 Use a white hanging buckle (835-610-42) as shown in the figure to organize the 3 wires connected to J4/J5/J1 together, and use one M5 screw (472-426-12) lock the white padlock to the screw hole in the TH corner.

The tool is a PH2 Phillips torque screwdriver, and the torque value is 1.6 N·m.

Note: The blue signal line is very fragile and should not be bent!